

Math 340: Elementary Matrix and Linear Algebra

PRACTICE FINAL

Name: _____ Section#: _____ NetID: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 120-minute exam. It consists of seven problems on eleven pages; the exam is six sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculator and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- **Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.**

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. Let C be a 3×3 matrix, and let

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 3 \\ 8 \\ 2 \end{bmatrix}$$

be the first, second, and third column of C , respectively.

(a) Compute the determinant of C .

(b) Is C invertible? If yes, what is the determinant of C^{-1} ? If no, explain why.

(c) Let P be the parallelepiped spanned by $\mathbf{u}_1, -2\mathbf{u}_2, 6\mathbf{u}_3$. What is the volume of P ?

(d) Is $\mathbf{v} = \begin{bmatrix} 4 \\ 14 \\ 3 \end{bmatrix}$ in $\text{span}\{\mathbf{u}_1, 3\mathbf{u}_2\}$?

(e) Do the vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ form a basis for $\mathbb{R}^3$?

2. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ be a matrix in $\mathbb{R}^{2 \times 2}$.

(a) Find the set W_1 of all $B \in \mathbb{R}^{2 \times 2}$ such that $AB = BA$.

(b) Let W_2 is a subspace in $\mathbb{R}^{2 \times 2}$ defined by

$$W_2 = \left\{ \begin{bmatrix} a & 0 \\ b & a-b \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

Find a basis for W_2 .

(c) What is the dimension of W_2 ?

3. Let $L: P_2 \rightarrow P_2$ be a linear transformation defined by

$$L(at^2 + bt + c) = (b - c)t^2 + (c - a)t + (a - b).$$

(a) Find $\dim \text{range } L$.

(b) Find a basis for $\ker L$.

4. Let E, G, H all be 3×4 matrices.

(a) Is it possible that E has column rank 4? Justify your answer.

(b) Is it possible that G has row rank 1? Justify your answer.

(c) Suppose that you compute the row echelon form of matrix H and it has the form

$$\begin{bmatrix} 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Find the solution(s) to $H\mathbf{x} = \mathbf{0}$, if any solutions exist. (Use x_1, x_2, x_3, x_4 for the unknowns of the linear system.)

(d) Let $f : \mathbb{R}^s \rightarrow \mathbb{R}^t$ be a matrix transformation defined by $f(\mathbf{u}) = E\mathbf{u}$.

- i. What are s and t ?
- ii. Is f a linear transformation?

5. Is the matrix

$$A = \begin{bmatrix} 1 & -5 & 0 \\ 1 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

diagonalizable? Justify your answer.

6. Let $L: P_1 \rightarrow \mathbb{R}^2$ be a linear transformation defined by

$$L(at + b) = \begin{bmatrix} 3b - a \\ 2a + b \end{bmatrix}.$$

Let $S_1 = (1, t)$ be the standard basis of P_1 , and T_1 be the standard basis of $\mathbb{R}^2$. Furthermore, let $S_2 = (3t + 5, t + 2)$ be another basis of P_1 and $T_2 = \left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right)$ be another basis for $\mathbb{R}^2$.

(a) Find $P_{S_1 \leftarrow S_2}$

(b) Find $P_{S_2 \leftarrow S_1}$

(c) What is the representation of L with respect to S_2 and T_2 ?

7. (3+3+4+5 points)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 7 \\ -2 \\ -1 \end{bmatrix}$$

be two vectors in $\mathbb{R}^3$.

(a) Find the distance between the points whose position vectors are $\mathbf{v}_1$ and $\mathbf{v}_2$.

(b) Let θ be the angle between $\mathbf{v}_1$ and $\mathbf{v}_2$.

Find $\cos \theta$.

(c) Let $\mathbf{w}_1 = \begin{bmatrix} 1 \\ 3 \\ a \end{bmatrix}$ and $\mathbf{w}_2 = \begin{bmatrix} 7 \\ a \\ -1 \end{bmatrix}$. Find a if $\mathbf{w}_1 \perp \mathbf{w}_2$.

(d) Let $\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ be vectors from $\mathbb{R}^3$, and define $\mathbf{a} \odot \mathbf{b}$ by

$$\mathbf{a} \odot \mathbf{b} = a_1 b_1 - a_2 b_2 + a_3 b_3.$$

Is the operation $\odot$ an inner product in $\mathbb{R}^3$?